

where $\mathbf{n}$ is the outward normal to ∂D . Denote P_L as the orthogonal projection of $L^2(D)^2$ onto H . The space of vector fields whose gradients are integrable in $L^2(D)^2$ is also relevant and we define

$$V := \{\mathbf{u} \in H^1(D)^2 : \nabla \cdot \mathbf{u} = 0, \mathbf{u}|_{\partial D} = 0\}.$$

We denote the norms associated to H and V respectively as $|\cdot|$ and $\|\cdot\|$.

The Stokes operator is defined as $Au = -P_L \Delta u$, for any vector field $\mathbf{u} \in V \cap H^2(D)^2$. Since A is self-adjoint with a compact inverse, we infer that A admits an increasing sequence of eigenvalues $\lambda_k \sim k$ diverging to infinity with the corresponding eigenvectors e_k forming a complete orthonormal basis for H . We denote by P_N and Q_N the projection onto $H_N = \text{span}\{e_k : k = 1, \dots, N\}$ and its orthogonal complement, respectively. Recall the generalized Poincaré inequalities

$$\|P_N \mathbf{u}\|^2 \leq \lambda_N |P_N \mathbf{u}|^2, \quad |Q_N \mathbf{u}|^2 \leq \lambda_N^{-1} \|Q_N \mathbf{u}\|^2 \quad (3.6)$$

hold for all sufficiently smooth $\mathbf{u}$ and any $N \geq 1$.

We make the following Hypotheses:

H₁. The function $\sigma = (\sigma_1, \dots, \sigma_m)$ is bounded and Lipschitz, i.e., there exist constants $B_0, L >$ such that

$$\begin{aligned} |\sigma(\mathbf{u})|^2 &= \sum_{k=1}^m |\sigma_k(\mathbf{u})|^2 \leq B_0, \text{ for all } \mathbf{u} \in H; \\ |\sigma(\mathbf{u}) - \sigma(\mathbf{v})|^2 &= \sum_{k=1}^m |\sigma_k(\mathbf{u}) - \sigma_k(\mathbf{v})|^2 \leq L |\mathbf{u} - \mathbf{v}|^2, \text{ for all } \mathbf{u}, \mathbf{v} \in H. \end{aligned}$$

H₂. There exists $N \in \mathbb{N}$ such that

$$P_N H \subset \text{Range}(\sigma(\mathbf{u})), \text{ for all } \mathbf{u} \in H.$$

Moreover, the corresponding pseudo-inverse operators $\sigma_k^{-1} : P_N H \rightarrow H$ are uniformly bounded, i.e., there exists a constant C_0 such that

$$|\sigma(\mathbf{u})^{-1}(P_N \mathbf{w})| \leq C_0 |P_N \mathbf{w}|, \text{ for all } \mathbf{u}, \mathbf{w} \in H.$$

H₃. Finally, we assume that N is sufficiently large such that

$$\lambda_N > \frac{L}{\nu} + \frac{C_D^2}{\nu^3} B_0. \quad (3.7)$$

The existence of a unique strong solution to equation (3.5) is standard, see e.g. [28]. Furthermore, as the energy estimates and Feller property can be obtained directly, we prove these two parts first.

Energy estimates

Proposition 3.3. *Assume H₁. Then*

$$\mathbb{E}|\mathbf{u}|^2 \leq e^{-2\nu t} |x|^2 + \frac{B_0}{2\nu}. \quad (3.8)$$